

Flexible DAS Implementation over a Self-Homodyne Optical Link

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Abstract We demonstrate a flexible ISAC architecture over a 25-km polarization-division-multiplexed self-homodyne link using a shared continuous frequency-swept carrier, achieving 160-Gbps 16QAM transmission with negligible communication penalty and 30-kHz vibration recovery. High-sampling-rate DAS is realized via continuous-wave sub-band multiplexing, increasing sampling rate 45× without spectral overhead. ©2026 The Author(s)

Introduction

Driven by cloud services and AI workloads, data centers and their interconnects (DCI) require optical links that are low-power, highly reliable, and observable. Self-homodyne detection (SHD) addresses the first two by sharing the transmitter laser and remotely delivering the local oscillator (LO), eliminating complex digital signal processing (DSP) blocks such as frequency-offset estimation and carrier-phase recovery [1]. Integrated sensing and communication (ISAC) addresses observability by reusing the same physical channel for link monitoring. Integrating ISAC into SHD links without sacrificing throughput or sensing performance is therefore a key challenge.

ISAC in optical links can be realized through forward or backward sensing, which are fundamentally complementary. Forward sensing, such as state-of-polarization (SOP) based polarization monitoring [2], [3], offers low-cost continuous monitoring but lacks accurate localization, whereas backward sensing provides precise localization at the cost of an extra channel (wavelength, timeslot, or orthogonal resource) [4], [5]. A practical ISAC architecture should ideally combine both.

A promising low-overhead route is to let a linear-frequency-modulated (LFM) carrier serve both functions [6], [7], particularly attractive in SHD systems, where the LFM LO naturally acts as the sensing probe. Recent SHD-LFM demonstrations have achieved data/LO delivery and optical frequency-domain reflectometry (OFDR) with negligible transmission penalty and 2-cm spatial resolution over 10-km multi-core fiber, through linear phase compensation [8], [9]. However, their sensing sampling rate is fundamentally limited by the LFM repetition period, which must exceed the fiber round-trip time (RTT), so the detectable acoustic bandwidth shrinks with distance, hindering high-frequency vibration monitoring over long-reach links. A communication-compatible ISAC architecture that unifies low-cost

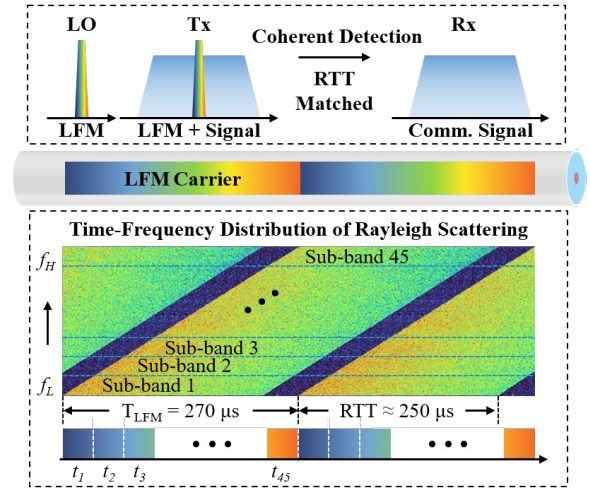


Fig. 1: Proposed ISAC working principle.

forward monitoring with precise, high-sampling-rate backward sensing is therefore still lacking.

We propose a flexible DAS architecture for SHD links, enabling on-demand switching between low-cost forward monitoring and precise backward sensing at high sampling rate. Over a 25-km, 160-Gbps 16QAM polarization-division-multiplexed (PDM) SHD link, a continuous LFM carrier simultaneously drives communication and ϕ -OTDR. At the sensing receiver, the backward Rayleigh-scattering signal is digitally partitioned into sub-bands with staggered time delays, raising the sampling rate by 45×. We demonstrate 30-kHz vibration recovery—15× the RTT-limited bandwidth—enabling robust ISAC for long-reach acoustic monitoring.

Backward Sensing ISAC Principle

Fig. 1 illustrates the proposed backward ISAC sensing architecture, in which a periodically swept continuous LFM carrier is shared by SHD and DAS. The polarization-multiplexed signal and the co-propagating LO originate from the same swept carrier and propagate over the same fibre. Their optical phases can be expressed as:

$$\varphi_s(t) = 2\pi f_0 t + \pi k t^2 \quad (1)$$

Reconfigurable Two-Stage Self-Homodyne Coherent Detection-ISAC Monitoring Platform

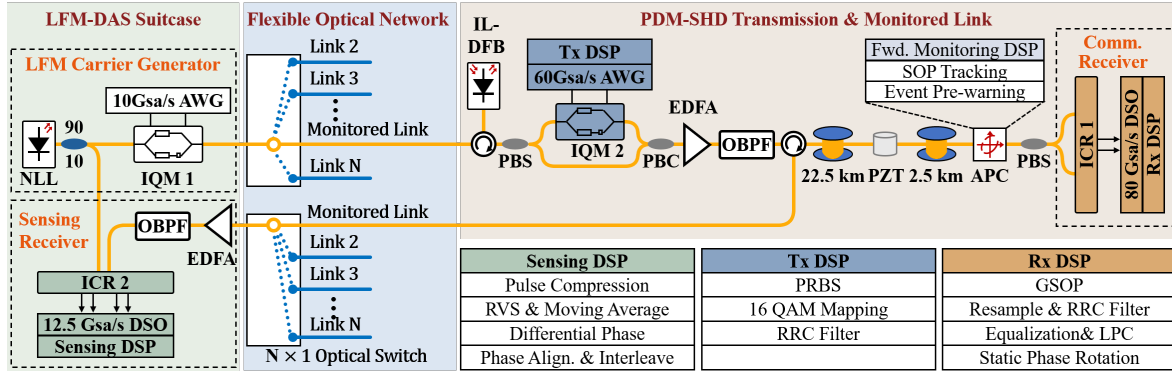


Fig. 2: Experimental setup of proposed ISAC scheme over self-coherent communication system.

$$\varphi_{LO}(t) = 2\pi f_0(t - \Delta t) + \pi k(t - \Delta t)^2 \quad (2)$$

which yield the coherent-beat phase [8], [9]:

$$\varphi_b(t) = 2\pi k\Delta t \cdot t + 2\pi f_0\Delta t - \pi k\Delta t^2 \quad (3)$$

where $f_0, k, \Delta t$ denote the initial optical frequency, the chirp rate of the LFM and relative-time delay (RTD) between the signal and LO, respectively. When RTD matching is satisfied ($\Delta t \approx 0$), the LFM-induced phase term is intrinsically suppressed, so no extra frequency-offset-estimation stage is required beyond conventional self-coherent detection in DSP.

For sensing, as shown in Fig. 1, the same 1–3.7 GHz, 270- μ s LFM carrier is reused as the DAS probe, and its backward Rayleigh-scattering signal is digitally partitioned in the frequency domain by matched-filter pulse compression into 45 receiver-side 60-MHz/6- μ s sub-bands. Each sub-band is independently demodulated to yield a slow-time sample. For sub-band bandwidth B_{sub} , the theoretical pulse-compressed spatial resolution [10] and effective DAS sampling rate f_s can be given as:

$$\Delta z \approx \frac{c}{2nB_{sub}} \quad (4)$$

$$f_s = \frac{k}{\Delta f_{step}} = \frac{10\text{MHz}/\mu\text{s}}{60\text{MHz}} \approx 167\text{ kSa/s} \quad (5)$$

where Δf_{step} represents center-frequency spacing of sub-bands. Therefore, the sub-band width can be flexibly configured according to the required sensing reach, spatial resolution, and measurement acoustic bandwidth. By employing this continuous-wave sub-band multiplexing scheme, communication and sensing share the same carrier with highly efficient spectral reuse, preserving self-coherent communication compatibility while enabling high-sampling-rate and long-reach DAS.

Experiment Setup

As shown in Fig. 2, the proposed architecture supports two operating modes sharing the same transmission fiber. In the default communication

mode, a distributed feedback (DFB) laser generates both the data-carrying signal and the co-propagating LO for SHD, while a receiver-side polarization controller performs forward polarization sensing. Upon detection of a link anomaly, the DFB is injection-locked by a remote LFM source, instantly converting it into an LFM carrier that drives φ -OTDR-based backward sensing over the same fiber. This dual-mode operation can be realized by a flexible optical network that reconfigures a few spare ports at the transmitter.

In the communication mode, a conventional PDM-SHD scheme was adopted. The DFB output was split by a 90:10 coupler: 90% served as the LO, and 10% was modulated with a 160-Gbps 16QAM waveform from a 60-GSa/s arbitrary waveform generator (AWG). The signal and LO were combined by a polarization beam combiner (PBC), amplified by an erbium-doped fiber amplifier (EDFA), and filtered to suppress out-of-band amplified spontaneous emission (ASE), yielding a carrier-to-signal power ratio (CSPR) of ~ 10.5 dB. The signal-LO RTD was matched to millimeter-level accuracy, giving a residual LFM-induced frequency offset of ~ 50 Hz at 10 MHz/ μ s—far below the SHD tolerance. At the receiver, a manual polarization controller (emulating an adaptive polarization controller, APC, in this proof-of-concept) separated the LO from the signal and served as the forward polarization sensor. The demultiplexed signal was detected by an integrated coherent receiver (ICR) and sampled by an 80-GSa/s digital storage oscilloscope (DSO) for offline DSP. A circulator after the transmitter-side EDFA forwards the signal and LO to the fiber, with its backward port idle in this mode.

For backward sensing, the backward port of the transmitter-side circulator is routed by the flexible optical network to a remote sensing module, reusing the existing communication-mode hardware. In this module, a narrow-linewidth laser (NLL) seeds an I-Q modulator driven by a 10-GSa/s AWG to generate a continuous 1–3.7 GHz,

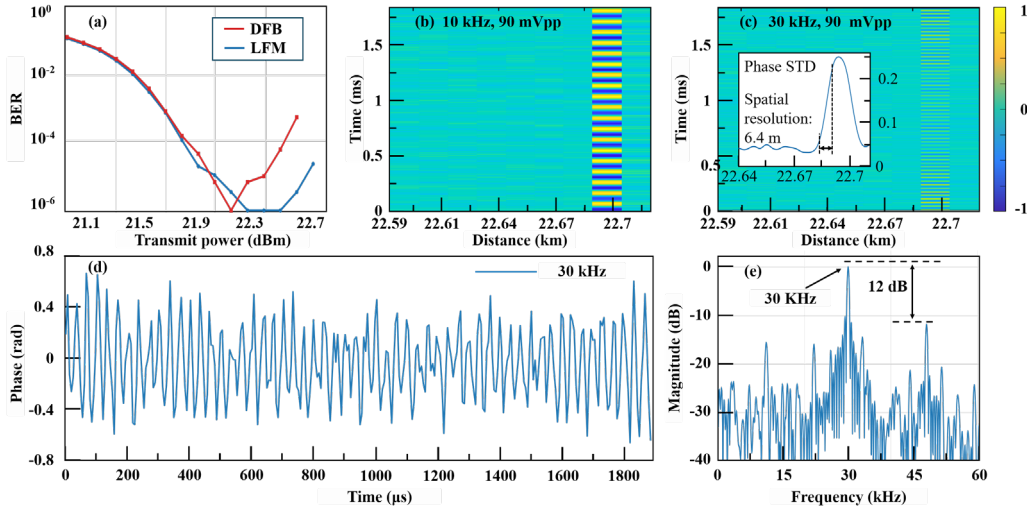


Fig. 3: Experimental results. (a) BER versus transmit power for DFB and LFM carriers. (b), (c) Interleaved phase maps of 10-kHz and 30-kHz vibrations (d) recovered 30-kHz time-domain trace and (e) corresponding FFT spectrum.

270- μ s single-sideband LFM waveform, while a 10% tap taken directly from the NLL serves as the local optical reference. Once routed into the communication link, this LFM waveform injection-locks the transmitter DFB, which coherently regenerates the swept waveform at constant output power with preserved frequency and phase, while suppressing the residual carrier, unwanted sideband, and spectral ripple. The ISAC link comprises a 22.5-km single-mode fiber (SMF), a 22-m lead zirconate titanate (PZT) vibration section, and a further 2.5-km SMF. Rayleigh backscattering returns through the same circulator to the sensing receiver, where it is coherently detected by ICR2 using the 10% NLL tap as the LO and sampled by a dedicated 12.5-GSa/s DSO.

The 1–3.7 GHz, 270- μ s swept carrier was partitioned into 45 sub-bands of 60 MHz / 6 μ s. Each sub-band was pulse-compressed by matched filtering, followed by intra-band masking, rotated-vector summation (RVS), and moving-average processing to suppress in-band interference from the co-transmitted communication signal and to mitigate Rayleigh-induced interference fading [11]. Differential-phase demodulation yielded the acoustic response within each sub-band, and inter-band phase alignment and interleaving across 45 sub-bands reconstructed a high-sampling-rate DAS sequence.

Results and discussions

Figure 3(a) shows the bit error rate (BER) of the SHD-received signal versus launched optical power after 25-km transmission. A BER below 1×10^{-6} is achieved at the optimum launch power, and the LFM-induced power penalty at 1×10^{-2} soft-decision forward error correction (SD-FEC) threshold is below 0.1 dB. Notably, at higher launch powers where the non-swept LO suffers from stimulated Brillouin scattering (SBS), the

LFM scheme even outperforms the non-swept case owing to its elevated SBS threshold, confirming the negligible impact of ISAC operation on communication performance.

For sensing, Figs. 3(b) and 3(c) show the recovered phase maps of the 10-kHz and 30-kHz vibrations, respectively. By interleaving 45 receiver-side sub-bands of 60 MHz, the effective sampling rate reaches 166.7 kSa/s over the 25-km link, 45 $\times$ the RTT-limited rate of ~ 4 kSa/s, extending the detectable vibration bandwidth from the corresponding Nyquist limit of ~ 2 kHz to beyond 30 kHz. A spatial resolution of 6.4 m is achieved through pulse compression, determined by the 20-MHz intra-sub-band bandwidth for RVS and the moving-average tap count. Both vibrations are recovered within this extended bandwidth. The time-domain trace and FFT spectrum of the 30-kHz case in Figs. 3(d) and 3(e) exhibit a distinct 30-kHz tone with ~ 12 dB peak SNR. The reduced phase amplitude at 30 kHz is attributed to the finite high-frequency response of the PZT and to partial signal cancellation in RVS, as the intra-sub-band phase variation deviates from the slow-varying assumption. Residual spurious harmonics are associated with the limited waveform fidelity of the 10-GSa/s AWG and can be mitigated in future implementations.

Conclusions

We present a flexible ISAC architecture for self-homodyne (SHD) links that supports on-demand switching between forward polarization monitoring and backward DAS on a shared LFM carrier. A continuous-wave sub-band multiplexing scheme decouples the DAS sampling rate from the fiber round-trip time, offering a communication-compatible, low overhead, and reconfigurable path toward observable SHD interconnects for future DCI networks.

Acknowledgements

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